Multi-Material CT Simulation and Synthetic Data Generation Using gVXR

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General Objective

The objective of this project is to investigate and develop a computational workflow for **generating synthetic X-ray projection and CT data using gVXR**, with particular emphasis on multi-material objects and realistic CT imaging conditions.

The project will use available 3D models, including **STL models of multi-material samples**, as input and investigate methods for defining object geometry, material properties, X-ray source and detector configurations, and other relevant simulation parameters.

The aim is to develop a reusable simulation module capable of generating synthetic CT data that may later support **CT reconstruction, algorithm development, validation, image processing, and AI-based applications** within the larger unified CT analysis framework.

The exact simulation workflow, methods, material definitions, and computational implementation are **not fixed initially** and should be selected after literature review and discussion.

Main Objectives

The project may investigate the following areas:

- Understanding the gVXR simulation framework and its capabilities.
- Importing and handling 3D object geometries, including STL files.
- Investigation of methods for representing **single-material and multi-material objects**.
- Assignment or definition of appropriate material properties for X-ray simulation.
- Configuration of X-ray source, object, and detector geometry.
- Investigation of relevant acquisition parameters for synthetic CT simulation.
- Generation of synthetic X-ray projection images.
- Generation of projections at multiple angular positions.
- Creation and organization of projection datasets suitable for CT reconstruction.

- Investigation of methods for generating or organizing sinogram data, where appropriate.
- Comparison between simulated data and available real experimental projection or CT data, where applicable.
- Investigation of how changes in geometry, material properties, source parameters, detector configuration, and acquisition parameters affect the generated projections.
- Investigation of methods for generating datasets with controlled variations.
- Investigation of noise or other realistic imaging effects, where supported or appropriate.
- Development of a structured synthetic dataset generation workflow that may support future reconstruction and AI projects.
- Preferably using **Python and/or C++** depending on compatibility with gVXR and performance requirements.
- Exploring C++ implementation or integration where practical and beneficial.
- Considering CPU and GPU computational capabilities according to the available gVXR implementation and computational requirements.
- Developing the work in a modular form so that it can later be integrated into the unified CT application.

The project is not limited to generating one fixed dataset. The broader aim is to investigate how a **reusable synthetic CT data generation pipeline** can be developed.

Initial Resources for Exploration

You may begin by exploring:

- gVXR documentation and example implementations.
- gVXR source code and available examples.
- X-ray image formation and attenuation principles.
- X-ray projection geometry.
- Cone-beam CT acquisition geometry.
- Multi-material X-ray simulation.

- STL and other 3D geometry formats.
- Material representation and X-ray attenuation properties.
- Synthetic projection generation.
- CT simulation workflows.
- Open-source CT simulation and reconstruction tools for comparison and understanding.
- Relevant literature on synthetic CT data generation and physics-based X-ray simulation.

The following resources may be provided for experimentation:

- Available STL models of real or multi-material samples.
- Available material information, where applicable.
- Selected real CT projection data.
- Selected reconstructed CT datasets.
- Available CT geometry and acquisition parameters.
- Nikon micro-CT metadata or HTML reports, where useful for understanding real acquisition conditions.

These are only **starting resources**. You are encouraged to identify and propose other suitable methods, libraries, tools, or datasets based on your initial study.

Prepare Your Plan of Action

Before starting the main implementation, please study the project area and prepare a brief **Plan of Action** describing:

1. Your understanding of the project and its objectives.
2. Your understanding of gVXR and its capabilities.
3. Relevant literature, software, libraries, examples, or existing implementations you have explored.
4. Your proposed approach for importing and handling the provided STL models.
5. How you propose to represent and simulate multi-material objects.

6. Your proposed approach for defining X-ray source, detector, and object geometry.
7. The simulation parameters you think should be investigated.
8. Your proposed workflow for generating projection data at multiple angles.
9. Whether and how you propose to generate or organize sinogram data.
10. How you propose to introduce controlled variation into the synthetic datasets.
11. Whether noise, detector effects, or other realistic imaging effects should be investigated.
12. How you propose to compare or validate simulated data against available real experimental data, where applicable.
13. Your proposed use of **Python, C++, or a combination of both**.
14. Your understanding of the available CPU and GPU capabilities relevant to the proposed workflow.
15. A preliminary idea of how the simulation pipeline or code could be structured.
16. The input data required for the simulation.
17. The expected outputs and data formats.
18. How you propose to test and validate the generated synthetic data.
19. Any technical challenges, assumptions, limitations, or questions you have identified.